

CHAPTER 5: MORPHOLOGY OF FLOWERING PLANTS

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1. The Root System

In the vast majority of dicotyledonous plants, the direct elongation of the embryonic radicle leads to the formation of the **primary root**, which penetrates the soil. This main axis branches into lateral roots of successive orders, commonly referred to as secondary, tertiary, and subsequent root lines. Together, the primary root and its lateral branches constitute the structural architecture of the **tap root system**, which is prominently typified by the mustard plant.

Conversely, in monocotyledonous plants, the primary root is inherently short-lived and is rapidly replaced by a dense cluster of numerous roots. These specialized roots originate directly from the base of the stem axis and form the **fibrous root system**, structurally characteristic of wheat. In unique modifications, roots emerge from structural zones of the plant body other than the radicle (such as in grass, *Monstera*, and the banyan tree); these are designated as **adventitious roots**.

Core Biological Functions of the Root System:

- Efficient absorption of water and essential mineral nutrients from the surrounding soil matrix.
- Providing secure structural and mechanical anchorage to the above-ground shoot system.
- Storage of primary reserve metabolic food materials.
- Biosynthesis of fundamental plant growth regulators and hormones.

Structural Regions of the Root Tip:

1. **Root Cap:** A protective, thimble-like anatomical structure covering the absolute apex, safeguarding the tender growing tip as it makes its structural way through soil layers.
2. **Region of Meristematic Activity:** Positioned a few millimeters above the root cap cover. The component cells are uniquely small, exceptionally thin-walled, contain dense protoplasm, and undergo repeated mitotic divisions.
3. **Region of Elongation:** Positioned proximal to the meristematic zone, where newly divided cells undergo rapid linear elongation and enlargement, driving the root's growth in length.
4. **Region of Maturation:** The zone proximal to the elongation region where cells gradually differentiate and reach specialized structural maturity. From the epidermal layer of this specific region, fine, delicate, thread-like extensions called **root hairs** develop to maximize water and mineral absorption surface area.

2. The Stem

The stem represents the ascending main developmental axis of the plant, emerging directly from the plumule of the embryo during seed germination processes. It bears branches, leaves, flowers, and fruits, and is

definitively distinguished from the root by the structural presence of **nodes** (the exact points where leaves are borne) and **internodes** (the explicit segments between two successive nodes).

Stems consistently possess buds that can be terminal (apical) or axillary in position. While young stems are typically green, herbaceous, and photosynthetic, they frequently become woody, robust, and dark brown with maturity. The primary function of the stem is spreading out branches to optimize light interception for leaves, and to systematically conduct water, minerals, and photosynthates. Certain stems are highly modified for specialized metabolic roles including food storage, physical support, mechanical protection, and vegetative propagation mechanisms.

3. The Leaf

The leaf is a lateral, typically flattened structural organ borne at the stem nodes. Leaves develop from the shoot apical meristems and are sequentially arranged in an **acropetal order**. Each leaf bears an axillary bud in its axil, which can subsequently develop into a lateral branch. Leaves are the primary vegetative organs for photosynthesis.

Anatomy of a Typical Leaf:

- **Leaf Base:** The segment attaching the leaf to the stem node, which may bear two small, lateral leaf-like structures called **stipules**. In monocots, the leaf base frequently expands into a sheath wrapping around the stem. In leguminous plants, it can become swollen, forming a structural platform called a **pulvinus**.
- **Petiole:** The long, flexible stalk extending the leaf blade to capture sunlight. It allows leaf blades to flutter gently in the wind, cooling the organ and refreshing the air layer at the surface.
- **Lamina (Leaf Blade):** The broad, green, expanded part containing a framework of veins and veinlets, centered around a prominent middle vein called the **midrib**. Veins provide crucial structural rigidity and serve as vascular channels for transport.

Venation Patterns:

The arrangement of veins within the lamina is termed venation. When veinlets branch out into a complex network, it is termed **reticulate venation** (prevalent in dicotyledons). When veins run strictly parallel to one another, it is called **parallel venation** (characteristic of most monocotyledons).

Leaf Classification:

- **Simple Leaf:** The lamina is completely entire, or if incised, the incisions do not reach the central midrib axis.
- **Compound Leaf:** The incisions extend fully to the midrib, partitioning the blade into individual leaflets. Bud presence is restricted to the axil of the main petiole and is never found in the axils of individual leaflets.
 - *Pinnately Compound:* Leaflets are arranged along a common central axis, the **rachis** (e.g., Neem).
 - *Palmately Compound:* Leaflets radiate outward from a single common point at the tip of the petiole (e.g., Silk Cotton).

Phyllotaxy:

Phyllotaxy defines the pattern of leaf arrangement on the stem axis, categorized into three principal types:

Type	Structural Arrangement	Key Examples
Alternate	A single leaf arises at each node in an alternating sequence.	China rose, Mustard, Sunflower
Opposite	A pair of leaves arises at each node, facing directly opposite each other.	Calotropis, Guava
Whorled	More than two leaves emerge at a single node to form a circular cluster.	Alstonia

4. The Inflorescence

A flower is fundamentally a modified shoot wherein the shoot apical meristem transforms into a specialized floral meristem. During this transformation, the internodes do not elongate, causing the axis to become highly condensed, and the apex generates successive floral appendages laterally rather than vegetative leaves. When a shoot tip undergoes absolute transformation into a flower, it occurs solubly (solitary).

The collective arrangement of flowers on the primary floral axis is termed the **inflorescence**. It is classified into two primary architectural types based on apical behavior:

- **Racemose Inflorescence:** The main axis retains its capacity for continuous growth, and flowers are produced laterally in a strict **acropetal succession**.
- **Cymose Inflorescence:** The main axis terminates abruptly in a flower, strictly limiting its linear growth. The flowers are consequently borne in a **basipetal order**.

5. The Flower

The flower is the specialized reproductive unit in angiosperms, structurally organized for sexual reproduction. A typical flower comprises four distinct whorls arranged sequentially on the swollen terminal end of the stalk (pedicel), known as the **thalamus** or **receptacle**.

The outer two whorls, the **calyx** (sepals) and **corolla** (petals), function as accessory structural organs. The inner two, the **androecium** (stamens) and **gynoecium** (carpels), are the primary reproductive organs. When calyx and corolla are visually indistinguishable, the outer envelope is termed a **perianth** (e.g., Lily).

Floral Symmetry & Metrics:

- **Actinomorphic (Radial):** Can be divided into identical halves along any radial plane passing vertically through the central axis (e.g., Mustard, Datura, Chilli).

- **Zygomorphic (Bilateral):** Can be divided into structural halves along only one single specific vertical plane (e.g., Pea, Gulmohur, Bean, Cassia).
- **Asymmetric (Irregular):** Cannot be divided into equivalent halves by any vertical plane passing through the center (e.g., Canna).
- **Whorl Counts:** Appendages can be trimerous, tetramerous, or pentamerous (multiples of 3, 4, or 5). Flowers possessing a reduced leaf at the base of the pedicel are termed *bracteate*; those lacking it are *ebracteate*.

Relative Position of Floral Parts on the Thalamus:

1. **Hypogynous:** The gynoecium is placed at the absolute highest point on the thalamus, while calyx, corolla, and androecium insert below it. The ovary is classified as **superior** (e.g., Mustard, China rose, Brinjal).
2. **Perigynous:** The gynoecium is positioned centrally, and the other floral whorls sit along the rim of a cup-like thalamus at roughly equivalent horizontal levels. The ovary is classified as **half-inferior** (e.g., Plum, Rose, Peach).
3. **Epigynous:** The margin of the thalamus grows vertically upward, completely engulfing the ovary wall and fusing seamlessly with it. The other floral components arise strictly above the ovary level, rendering the ovary **inferior** (e.g., Guava, Cucumber, ray florets of Sunflower).

Aestivation Types:

Aestivation defines the mode of arrangement of sepals or petals within a floral bud relative to other members of the same whorl:

- **Valvate:** Structural margins merely touch one another without any overlapping (e.g., *Calotropis*).
- **Twisted:** One margin overlaps the adjacent appendage, which in turn is overlapped by the next (e.g., China rose, Lady's finger, Cotton).
- **Imbricate:** Margins overlap each other unevenly, without a fixed, directional pattern (e.g., *Cassia*, Gulmohur).
- **Vexillary (Papilionaceous):** Comprises five distinct petals: a large posterior petal (standard) overlaps two lateral petals (wings), which enclose two smaller anterior fused petals (keel) (e.g., Pea, Bean).

Androecium & Gynoecium Configurations:

Stamens can be free (polyandrous) or united into distinct structural bundles: **monoadelphous** (one single bundle, e.g., China rose), **diadelphous** (two bundles, e.g., Pea), or **polyadelphous** (multiple bundles, e.g., Citrus). If stamens adhere to petals, they are epipetalous (e.g., Brinjal). Carpels within a gynoecium are termed **apocarpous** if they remain completely free from one another (e.g., Lotus, Rose) and **syncarpous** if they are structurally fused together (e.g., Mustard, Tomato).

Placentation Typologies:

Placentation describes the specific arrangement of ovules within the ovary chamber:

- **Marginal:** The placenta forms a longitudinal ridge along the ventral suture, bearing rows of ovules (e.g., Pea).
- **Axile:** The placenta is positioned along the central axis of a multilocular, septate ovary (e.g., China rose, Tomato, Lemon).
- **Parietal:** Ovules develop along the inner peripheral wall of a unilocular ovary, which may become false-septate into two chambers (e.g., Mustard, *Argemone*).
- **Free Central:** Ovules are borne directly on a central axis completely devoid of dividing septa (e.g., *Dianthus*, Primrose).
- **Basal:** The placenta develops directly at the base of the ovary, supporting a single ovule (e.g., Sunflower, Marigold).

6. Fruits and Seeds

The fruit represents a post-fertilization modification of the mature or ripened ovary. A fruit developing completely independent of fertilization is termed **parthenocarpic**. In structural composition, a fruit consists of an outer wall, the **pericarp** (which may differentiate into an outer thin *epicarp*, a middle *mesocarp*, and an inner *endocarp*), enclosing the seeds. In mango and coconut, the fruit is structurally defined as a **drupe**—developing from monocarpellary, superior ovaries and yielding a single seed. The mango features a fleshy, edible mesocarp, whereas the coconut contains a highly fibrous mesocarp.

Following fertilization, ovules transform into seeds, each consisting of a protective seed coat (outer *testa* and inner *tegmen*) and an embryo. Dicotyledonous seeds possess two cotyledons flanking an embryonal axis with a radicle and plumule, and may lack an endosperm at maturity (non-endospermous, e.g., bean, gram, pea). Monocotyledonous seeds (e.g., maize) typically feature a dominant, food-storing **endosperm** bounded by a proteinaceous **aleurone layer**, a single shield-shaped cotyledon called a **scutellum**, and protective embryonic sheaths: the **coleoptile** (protecting the plumule) and **coleorhiza** (protecting the radicle).

7. Taxonomy: The Solanaceae Family

Commonly designated as the 'potato family', Solanaceae is a large, diverse family widely distributed across tropical, subtropical, and temperate geographic zones.

Taxonomic Summary:

- **Habit:** Mostly herbs, shrubs, and rare small trees.
- **Stem:** Herbaceous or rarely woody, aerial, erect, cylindrical, solid or hollow, hairy or glabrous; specialized underground tuberous stem in potato (*Solanum tuberosum*).
- **Leaves:** Alternate, simple, rarely pinnately compound, exstipulate; possessing distinct reticulate venation.
- **Inflorescence:** Solitary, axillary, or organized into a cymose structure as in *Solanum*.

- **Flower:** Bisexual, actinomorphic.
- **Calyx:** Sepals five, united (gamosepalous), persistent, showing valvate aestivation.
- **Corolla:** Petals five, united (gamopetalous), showing valvate aestivation.
- **Androecium:** Stamens five, distinct, alternating with petals and distinctly **epipetalous**.
- **Gynoecium:** Bicarpellary, obligately tilted, syncarpous; ovary superior, bilocular, containing a highly swollen placenta bearing numerous ovules with axile placentation.
- **Fruits & Seeds:** Berry or capsule; containing numerous endospermous seeds.

Standard Floral Formula

$$\oplus \quad \sigma \quad \varphi \quad K_{(5)} \quad \overline{C_{(5)} A_5} \quad G_{(2)}$$

Interpretation: Actinomorphic symmetry ($\oplus$), Bisexual ($\sigma \varphi$), 5 united sepals ($K_{(5)}$), 5 united petals ($C_{(5)}$) epipetalously fused with 5 stamens (A_5), bicarpellary syncarpous superior ovary ($G_{(2)}$).

Economic Importance:

- **Food Sources:** Tomato, Brinjal, Potato.
- **Spices:** Chilli.
- **Medicinals:** Belladonna, Ashwagandha.
- **Fumigatories:** Tobacco.
- **Ornamentals:** Petunia.